

Shizhe Liu

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EDUCATION

University of Oxford, Department of Computer Science
Computer Science and Philosophy (MCompSciPhil)

Oct 2024 - Present
Oxford

King Edward VI Camp Hill School for Boys

A-Level Further Maths (A), Maths (A*), Computer Science (A*), Physics (A*) and EPQ (A*)*

Sep 2022 - Jun 2024
Birmingham

EXPERIENCE

Independent Researcher

Jul 2025 - Present

Generative AI, Affective Computing, Digital Human and Human-Agent Interaction Research

- Developed **ResDiffFRG**, a novel reformulation of the diffusion process for facial reaction generation, improving facial-reaction correlation by 43% over existing baselines; currently under review at a top ML conference
- Developed **SAFRG**, a novel framework that aligns facial reaction generation with the intended verbal response; published in **Neurocomputing**
- Designed and evaluated transformer-based diffusion denoisers in PyTorch using multimodal representations

BT Applied Research Department

Applied research work experience

Jun 2022 - Jul 2022
Adastral Park, Ipswich

- Developed a real-time hand-gesture classification model for Microsoft HoloLens

PUBLICATIONS

1. **S. Liu**, J. Gu, X. Kong, S. Song. ResDiffFRG: Residual Diffusion for Multiple Appropriate Facial Reaction Generation. arXiv preprint. 2026. <https://doi.org/10.48550/arXiv.2609.33749>
2. **S. Liu**, X. Kong, J. Long, J. Gu, S. Song. SAFRG: Speech Aligned Multiple Appropriate Facial Reaction Generation. *Neurocomputing*, Elsevier, Volume 702, Article 134546. 2026. <https://doi.org/10.1016/j.neucom.2026.134546>
3. X. Kong, H. Zhu, H. Sun, Z. Guo, J. Gu, X. Ni, W. Zhang, **S. Liu**, S. Song. Learning Personalised Human Internal Cognition from External Expressive Behaviours for Real Personality Recognition. *The 40th Annual AAAI Conference on Artificial Intelligence (AAAI-26)*. 2026. <https://doi.org/10.1609/aaai.v40i3.37167>
4. **S. Liu**, S. Hapuarachchi, P. Shrikaanth, W. Shi, J. Regi. Applications of the XGBoost Machine Learning Algorithm in Particle Physics. In: Real Particle Physics Analysis by UK Secondary School Students using the ATLAS Open Data: an Illustration through a Collection of Original Student Research. *The European Physical Journal Plus*, Springer, Volume 139, Article 781, Section 9. 2024. <https://doi.org/10.1140/epjp/s13360-024-05518-z>

PROJECTS (SELECTED)

Oriel College JCR AI Assistant & Automation Tools

Jul 2025 - Oct 2025

Oriel College, University of Oxford

- Built and deployed an AI Chatbot to answer questions from freshers using Streamlit and a RAG pipeline
- Developed a live, automated Instagram intro generator, using Google APIs, Pillow and GitHub Actions

Finding Rare Particle Decays using XGBoost

Jan 2023 - Jun 2023

IRIS Best Research Project Award

- Developed an XGBoost-based approach for identifying rare particle-decay events in ATLAS Open Data; this work was subsequently published in **European Physical Journal Plus**

SKILLS & AWARDS

Technical: Python, PyTorch, NumPy, Scala, Haskell | Diffusion Models, Transformers, Generative Modelling, RAG

Awards: IRIS Best Research Project Award (2023) | British Mathematical Olympiad Distinction (2023) | Oxford University Computing Challenge Distinction (2023)